

BACHELOR OF COMPUTER APPLICATIONS
SEMESTER III
DBMS

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LIBRARY MANAGEMENT SYSTEM

Database Architecture & Design Specification

1. Name and Purpose of the Application

Application Name: LibSphere – Library Management System

Purpose

LibSphere is a computerized library management application developed to simplify and automate major library activities. The main objectives of the system are:

- **Book Inventory Management:** Maintain organized information about books, categories, authors, and the number of available copies.
- **Member Management:** Store and manage registered member details, contact information, and membership status.
- **Borrowing Management:** Record book issues, due dates, returns, and currently active loans.
- **Improved Efficiency:** Reduce dependency on manual registers, minimize book-tracking errors, and maintain consistency and accuracy of library records.

2. Required Tables

The LibSphere database is designed using six main relational tables:

1. Categories: Maintains the different genres, subjects, or classifications of books, such as Fiction, History, and Computer Science.

2. Authors: Stores details about authors whose books are available in the library.

3. Books: Contains information about book titles, ISBN numbers, publication details, and inventory quantities.

4. Book_Authors: A junction table used to establish the many-to-many relationship between books and authors.

5. Members: Stores information about registered library users, including their contact details and membership status.

6. Loans: Maintains records of books issued to members, including issue dates, due dates, return dates, and loan status.

3. Columns / Attributes for Each Table

Categories Table

The Categories table is used to organize books according to their subject, genre, or classification.

Column Name	Data Type	Constraints
category_id	INT	PRIMARY KEY, AUTO_INCREMENT
category_name	VARCHAR(50)	NOT NULL, UNIQUE

Authors Table

The Authors table contains basic information about the authors associated with books in the library catalog.

Column Name	Data Type	Constraints
author_id	INT	PRIMARY KEY, AUTO_INCREMENT
first_name	VARCHAR(50)	NOT NULL
last_name	VARCHAR(50)	NOT NULL
bio	TEXT	NULLABLE

Books Table

The Books table stores bibliographic information and keeps track of the available inventory of each book title.

Column Name	Data Type	Constraints
book_id	INT	PRIMARY KEY, AUTO_INCREMENT
title	VARCHAR(150)	NOT NULL
isbn	VARCHAR(13)	NOT NULL, UNIQUE
category_id	INT	FOREIGN KEY
publication_year	INT	NOT NULL
total_copies	INT	NOT NULL, DEFAULT 1
available_copies	INT	NOT NULL, DEFAULT 1

The category_id column establishes a relationship between the Books and Categories tables.

Book_Authors Table

The Book_Authors table acts as an intermediate or junction table. It allows a book to have multiple authors and also allows an author to be associated with multiple books.

Column Name	Data Type	Constraints
book_id	INT	PRIMARY KEY, FOREIGN KEY
author_id	INT	PRIMARY KEY, FOREIGN KEY

The combination of book_id and author_id forms a composite primary key.

Members Table

The Members table stores the personal and membership-related information of registered library users.

Column Name	Data Type	Constraints
member_id	INT	PRIMARY KEY, AUTO_INCREMENT
first_name	VARCHAR(50)	NOT NULL
last_name	VARCHAR(50)	NOT NULL
email	VARCHAR(100)	NOT NULL, UNIQUE
phone	VARCHAR(20)	NOT NULL
membership_date	DATE	NOT NULL
status	VARCHAR(20)	NOT NULL, DEFAULT 'Active'

Loans Table

The Loans table records every borrowing transaction performed by library members.

Column Name	Data Type	Constraints
loan_id	INT	PRIMARY KEY, AUTO_INCREMENT
book_id	INT	FOREIGN KEY
member_id	INT	FOREIGN KEY
loan_date	DATE	NOT NULL
due_date	DATE	NOT NULL
return_date	DATE	NULLABLE
status	VARCHAR(20)	NOT NULL, DEFAULT 'Issued'

The book_id and member_id columns connect each loan transaction to the corresponding book and library member.

4. Relationships Between Tables

Categories → Books: One-to-Many: A single category can be associated with several books, while each book is assigned to one category. This relationship is maintained through Books.category_id.

Books ↔ Authors: Many-to-Many: A book may have one or more authors, and an author may contribute to several books. The Book_Authors table is used to manage this many-to-many relationship.

Members → Loans: One-to-Many: One member can borrow books multiple times during their membership. However, each loan transaction is associated with only one member through Loans.member_id.

Books → Loans: One-to-Many: A particular book title can be borrowed repeatedly by different members over time. Each individual borrowing transaction refers to one book through Loans.book_id.

5. Overall Explanation

The LibSphere database follows Third Normal Form (3NF) principles. The design separates different types of information into individual tables to reduce duplication and maintain data consistency.

1. Normalized Database Structure: Authors, categories, books, members, and loan transactions are maintained separately. This avoids duplicate information and ensures that updates can be made in one appropriate location.

2. Efficient Author Management: The Book_Authors junction table supports books with multiple authors as well as authors who have written multiple books. This provides flexibility without adding unnecessary columns to the Books table.

3. Inventory and Loan Management: The `available_copies` attribute in the Books table indicates how many copies can currently be issued. When a book is borrowed, the application decreases the available count. When the book is returned, the available count is increased.

4. Maintaining Data Integrity: Primary keys uniquely identify records, while foreign keys maintain valid relationships between tables. Referential integrity rules help prevent inappropriate deletion of books or members that are associated with existing loan transactions.

Conclusion

The LibSphere database provides a structured and reliable foundation for managing library resources. By using normalized tables, primary keys, foreign keys, and appropriate relationships, the system can efficiently handle book cataloging, author management, member registration, inventory tracking, and borrowing/return operations.